

DevConnect — Graph Database Take-Home

A small graph-backed developer discovery application built for the Wexa AI CognoDB assignment.

Stack

- React + Vite frontend
- Node.js + Express backend
- Official Neo4j JavaScript driver
- CognoDB Cloud using Bolt/openCypher

Why a graph database?

DevConnect is centered on relationships: developers have skills, work on projects, projects use technologies, and developers know other developers. Questions such as "which skills belong to developers I know?" require graph traversal and are more natural to express as connected patterns than as repeated relational joins.

Graph model

- Developer
- Skill
- Project
- Technology

Relationships:

- Developer -[:HAS_SKILL]-> Skill
- Developer -[:WORKED_ON]-> Project
- Project -[:USES]-> Technology
- Developer -[:KNOWS]-> Developer

Multi-hop query

The application exposes a network view based on:

```
Developer -[:KNOWS]-> Developer -[:HAS_SKILL]-> Skill
```

This is a two-hop traversal used to discover skills through a developer's direct connections.

Run locally

Backend

```
cd backend
npm install
cp .env.example .env
```

```
# Add your CognoDB values to .env
npm run seed
npm run dev
```

Backend runs at <http://localhost:5000>.

Frontend

```
cd frontend
npm install
cp .env.example .env
npm run dev
```

Frontend normally runs at <http://localhost:5173>.

Environment variables

Backend:

- [COGNODB_URI](#)
- [COGNODB_USERNAME](#)
- [COGNODB_PASSWORD](#)
- [PORT](#)

Frontend:

- [VITE_API_URL](#)

Do not commit [.env](#) files.